Sigmoid function, $f(x) = \frac{1}{1+e^{-x}}$

Let $y = f(x)$

$$\therefore y = \frac{1}{1+e^{-x}}$$

1st derivative:

$$y = (1+e^{-x})^{-1}$$

$$\frac{dy}{dx} = -1 \cdot (1+e^{-x})^{-2} \cdot d/dx(1+e^{-x})$$

Now

$$\frac{d}{dx}(1+e^{-x}) = -e^{-x}$$

$$\frac{dy}{dx} = -(1+e^{-x})^{-2}(-e^{-x})$$

$$\frac{dy}{dx} = \frac{e^{-x}}{(1+e^{-x})^2}$$

Since

$$y = \frac{1}{1+e^{-x}}$$

$$1-y = 1-\frac{1}{1+e^{-x}}$$

$$1-y = \frac{e^{-x}}{1+e^{-x}}$$

$$y(1-y) = \frac{1}{1+e^{-x}} \times \frac{e^{-x}}{1+e^{-x}}$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$

$$y' = y(1-y)$$

$$\therefore f'(x) = f(x)\left(1-f(x)\right)$$